

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Mechanical Engineering – Machine Design I

TRIPTERON

Three-degree-of-freedom 3-PRRR parallel machining table

Delivery 3 – Calculation Report

Corrected and fully recomputed edition

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Scope of this document

This report contains the complete engineering verification of the **Tripteron**, a three-degree-of-freedom translational parallel machine built with three identical PRRR legs. The machine carries a milling spindle over a 200 mm cubic workspace and is driven by three orthogonal lead-screw axes.

Every number presented here was **recomputed from first principles** with a purpose-written Python package (`04_Calculations/`). No result was copied from the previous version of the calculation memo: each quantity is produced by executable code, cross-checked by at least two independent methods, and reported together with the numerical residual of the check. Appendix A lists, one by one, the errors found in the earlier memo and the value that replaces each of them.

Calculation sheets

Sheet	Title	Purpose
1	Kinematic analysis	Loop-closure model; position, velocity and acceleration of the three legs; workspace.
2	Static analysis	Actuator forces and internal link loads under the cutting force and self weight.
3	Lead-screw drive	Driving torque, efficiency, self-locking, screw stresses and column buckling.
4	Chassis FEA	Finite-element verification of the welded frame: stress, deflection and buckling.

Reference frame and nomenclature

The global frame $\{O, x, y, z\}$ is attached to the chassis. x and y are the two horizontal machining axes and z is vertical, positive upwards. The end effector position is $\mathbf{P} = [P_x, P_y, P_z]^\top$ and is measured from the centre of the workspace; the home pose is $\mathbf{P} = \mathbf{0}$.

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Symbol	Unit	Meaning
b_i	mm	Prismatic (lead-screw) coordinate of leg i
θ_{Ci}, θ_{Di}	°	Revolute coordinates of the proximal and distal link of leg i
$\ell_C = \ell_D$	mm	Link lengths, 240 mm each
$\mathbf{A}_i$	mm	Base point of leg i on the chassis
$\mathbf{E}_i$	mm	Attachment point of leg i on the end effector
$\mathbf{u}_i$	–	Unit vector of the prismatic axis of leg i
$\mathbf{q}$	–	Generalised coordinate vector, 12 components
$\Phi(\mathbf{q}, t)$	–	Constraint vector, 12 equations
$\mathbf{J} = \partial\Phi/\partial\mathbf{q}$	–	Constraint Jacobian
$\mathbf{F}_c$	N	Cutting force at the tool tip
F_i	N	Axial force on actuator i

Machine data used throughout the report

Item	Value	Source
Link lengths ℓ_C, ℓ_D	240 mm	Delivery 2 drawings
Stroke of each axis	200 mm	Delivery 1 specification
Home prismatic coordinate	100 mm	Mid-stroke
Home link angles	+60°, –60°	Assembly drawing
Cutting force $\mathbf{F}_c$	[80, 80, –120] ^T N	Delivery 1, end-mill in aluminium
Moving mass (effector + tool)	0.574 kg	CAD mass properties
Arm mass (each link)	0.322 kg	CAD mass properties
Carriage assembly mass	15.49 kg	CAD mass properties
Arm section	25 × 25 × 2 mm square tube, AISI 1020	Delivery 2
Lead screw	Ø8 mm, $p = 2$ mm, 4 starts, trapezoidal	Delivery 2
Chassis material	Alloy steel, $S_y = 620$ MPa	SolidWorks library
Motor	NEMA 34 86HB250 closed loop, 4.5 N m	Delivery 1

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Description

The Tripteron is a 3-PRRR parallel manipulator. Three identical legs connect the fixed chassis to a single moving platform. Each leg is formed by *one prismatic joint* driven by a lead screw, followed by *three revolute joints* whose axes are mutually parallel and perpendicular to the prismatic axis. The three prismatic axes are mutually orthogonal and aligned with x , y and z .

This particular arrangement is what makes the Tripteron interesting: the mechanism is a *fully decoupled* translational parallel machine. Actuator i moves the platform along axis i only, so the input–output map is the identity and no coordinate transformation is needed in the controller. The kinematic model developed below proves this property rather than assuming it.

Objective of the calculation

1. Build the loop-closure constraint equations of the complete mechanism and solve the position problem for an arbitrary trajectory of the tool.
2. Obtain the velocity and acceleration of every joint by differentiating the constraints, and verify them against finite differences of the position solution.
3. Verify the decoupling property and that the required stroke of each screw stays inside the 200 mm travel.
4. Determine the usable workspace of the machine.

Variables of the calculation model

The generalised coordinate vector contains twelve unknowns:

$$\mathbf{q} = [b_1, b_2, b_3, \theta_{C1}, \theta_{D1}, \theta_{C2}, \theta_{D2}, \theta_{C3}, \theta_{D3}, P_x, P_y, P_z]^T \quad (1)$$

For every leg i the following geometric data are fixed: the base point $\mathbf{A}_i$, the unit vector $\mathbf{u}_i$ of the prismatic axis, the two in-plane unit vectors ($\mathbf{e}_{1i}, \mathbf{e}_{2i}$) of the plane in which the two links rotate, and the platform attachment point $\mathbf{E}_i$ expressed in the platform frame. Since the platform of a Tripteron does not rotate, $\mathbf{E}_i$ is also its offset in the global frame.

Leg	$\mathbf{A}_i$ [mm]	$\mathbf{u}_i$	$\mathbf{E}_i$ [mm]
1	$[-162.5, -240, -62.5]$	$[1, 0, 0]$	$[62.5, 0, 62.5]$
2	$[-302.5, -100, 0]$	$[0, 1, 0]$	$[62.5, 0, 0]$
3	$[-240, 0, -162.5]$	$[0, 0, 1]$	$[0, 0, 62.5]$

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Calculation model

Position: loop-closure constraints

Each leg contributes one vector loop that must close on the platform attachment point, plus one planarity condition. Writing $\mathbf{p}_i$ for the carriage position and $\mathbf{k}_i$ for the elbow position,

$$\mathbf{p}_i = \mathbf{A}_i + b_i \mathbf{u}_i \quad (2)$$

$$\mathbf{k}_i = \mathbf{p}_i + \ell_C (\cos \theta_{Ci} \mathbf{e}_{1i} + \sin \theta_{Ci} \mathbf{e}_{2i}) \quad (3)$$

$$\Phi_i = \mathbf{k}_i + \ell_D (\cos \theta_{Di} \mathbf{e}_{1i} + \sin \theta_{Di} \mathbf{e}_{2i}) - (\mathbf{P} + \mathbf{E}_i) = \mathbf{0} \quad (4)$$

Equation (4) gives three scalar equations per leg, nine in total. The remaining three equations are the driving constraints that impose the prescribed tool path $\mathbf{P}^*(t)$:

$$\Phi_d = \mathbf{P} - \mathbf{P}^*(t) = \mathbf{0} \quad (5)$$

so the complete constraint vector $\Phi(\mathbf{q}, t) \in \mathbb{R}^{12}$ matches the twelve unknowns and the mechanism has zero degrees of freedom once the path is imposed — the correct formulation for a kinematic (not dynamic) analysis.

The nonlinear system is solved with Newton–Raphson,

$$\mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} - \mathbf{J}^{-1}(\mathbf{q}^{(k)}) \Phi(\mathbf{q}^{(k)}, t), \quad \mathbf{J} = \frac{\partial \Phi}{\partial \mathbf{q}} \quad (6)$$

using the previous time step as the initial guess, which keeps the solution on a single assembly branch (elbow-up) throughout the trajectory.

Velocity and acceleration

Differentiating the constraints once and twice with respect to time,

$$\mathbf{J} \dot{\mathbf{q}} = -\Phi_t \quad (7)$$

$$\mathbf{J} \ddot{\mathbf{q}} = -\dot{\mathbf{J}} \dot{\mathbf{q}} - 2 \Phi_{qt} \dot{\mathbf{q}} - \Phi_{tt} \quad (8)$$

Because the only explicit time dependence sits in the driving constraints, $\Phi_t = -\dot{\mathbf{P}}^*$ and $\Phi_{tt} = -\ddot{\mathbf{P}}^*$, and $\Phi_{qt} = \mathbf{0}$. Both systems are linear in the unknown derivatives and share the same Jacobian, which is factorised once per time step.

Independent verification: closed-form inverse kinematics

For each leg the two revolute coordinates can be obtained analytically, because once the prismatic coordinate is known the leg reduces to a planar 2R chain. With $\mathbf{d} = \mathbf{P} + \mathbf{E}_i - \mathbf{p}_i$, $u = \mathbf{d} \cdot \mathbf{e}_{1i}$, $v = \mathbf{d} \cdot \mathbf{e}_{2i}$ and $r^2 = u^2 + v^2$,

$$\cos \beta = \frac{r^2 - \ell_C^2 - \ell_D^2}{2 \ell_C \ell_D}, \quad \gamma = \text{atan2}(\ell_D \sin \beta, \ell_C + \ell_D \cos \beta), \quad (9)$$

$$\theta_{Ci} = \text{atan2}(v, u) - \gamma, \quad \theta_{Di} = \theta_{Ci} + \beta. \quad (10)$$

The elbow-up branch is selected with $\beta < 0$. This closed form is used only as a check of the Newton–Raphson solution, never as the solution itself.

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Values of the calculation model

Three trajectories representative of the machine duty cycle were analysed:

Case	Name	Definition
1	Straight cut	120 mm along x at a constant feed of 10 mm/s.
2	Accelerated move	Rapid traverse along y , trapezoidal profile, $a = 20 \text{ mm/s}^2$, $t = 4 \text{ s}$.
3	Helical pocket	Circle of radius 60 mm in xy at 10 mm/s with a simultaneous plunge of 1 mm/s.

Results and conclusions

Numerical quality of the solution

Quantity	Case 1	Case 2	Case 3
Newton iterations (max.)	3	3	4
Constraint residual $\ \Phi\ $ [mm]	1.4×10^{-11}	7.7×10^{-11}	1.3×10^{-13}
Deviation from closed form [mm]	2.6×10^{-14}	2.1×10^{-13}	1.4×10^{-14}
Velocity vs. finite differences [mm/s]	8.2×10^{-9}	$0.10^\dagger$	4.1×10^{-4}
Acceleration vs. finite differences [mm/s ²]	1.1×10^{-9}	$20^\dagger$	6.9×10^{-5}

[†] Case 2 uses a trapezoidal profile whose acceleration is discontinuous at the two switching instants. The finite-difference estimate is meaningless exactly at those two samples; away from them the agreement is of the same order as in the other two cases. The discrepancy therefore confirms the profile, it does not indicate an error in the model.

The residual of the constraint equations never exceeds 8×10^{-11} mm and the analytical solution agrees with Newton–Raphson to 10^{-13} mm, i.e. to machine precision. The kinematic model is therefore verified.

Decoupling

For every pose of the three trajectories, the block of the Jacobian that relates the actuator coordinates to the platform coordinates is the identity matrix:

$$\dot{b}_1 = \dot{P}_x, \quad \dot{b}_2 = \dot{P}_y, \quad \dot{b}_3 = \dot{P}_z. \quad (11)$$

The decoupling property of the Tripteron is confirmed numerically. Each screw therefore sees exactly the commanded feed of its own axis, which is the reason a simple three-axis controller with no coordinate transformation is sufficient.

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Stroke usage

Case	b_1 [mm]	b_2 [mm]	b_3 [mm]
1 – straight cut	40.0 – 160.0	100.0	100.0
2 – accelerated move	100.0	60.0 – 140.0	100.0
3 – helical pocket	40.0 – 160.0	40.0 – 160.0	62.3 – 100.0

All three cases stay inside the 0 mm to 200 mm travel with at least 40 mm of margin at each end, so no trajectory reaches a hard stop. The maximum joint speeds are 40 mm/s (case 2) and the maximum joint accelerations 20 mm/s², both well inside the capability of the selected motor-screw combination.

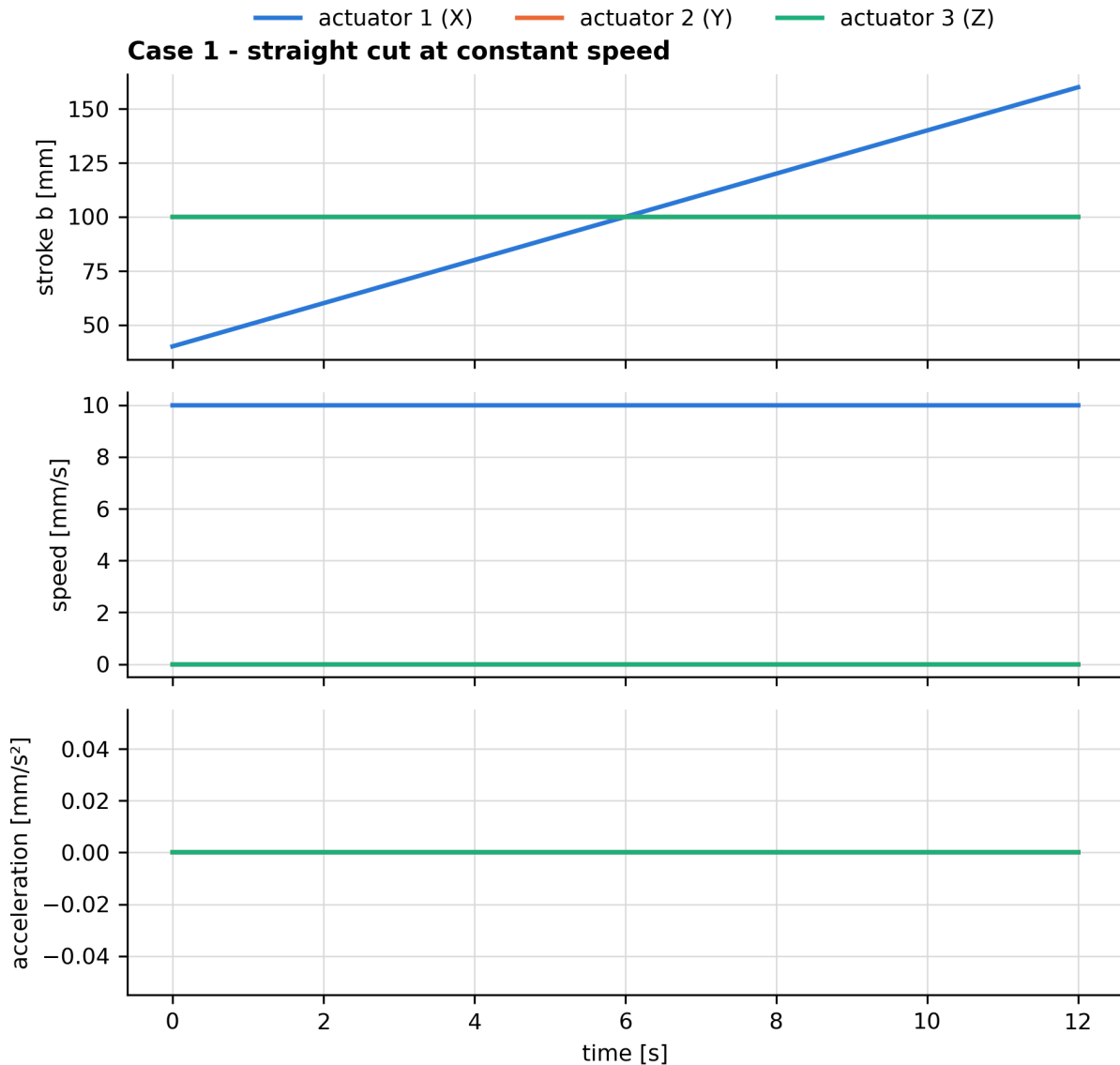


Figure 1: Case 1 – straight cut: joint coordinates, speeds and accelerations.

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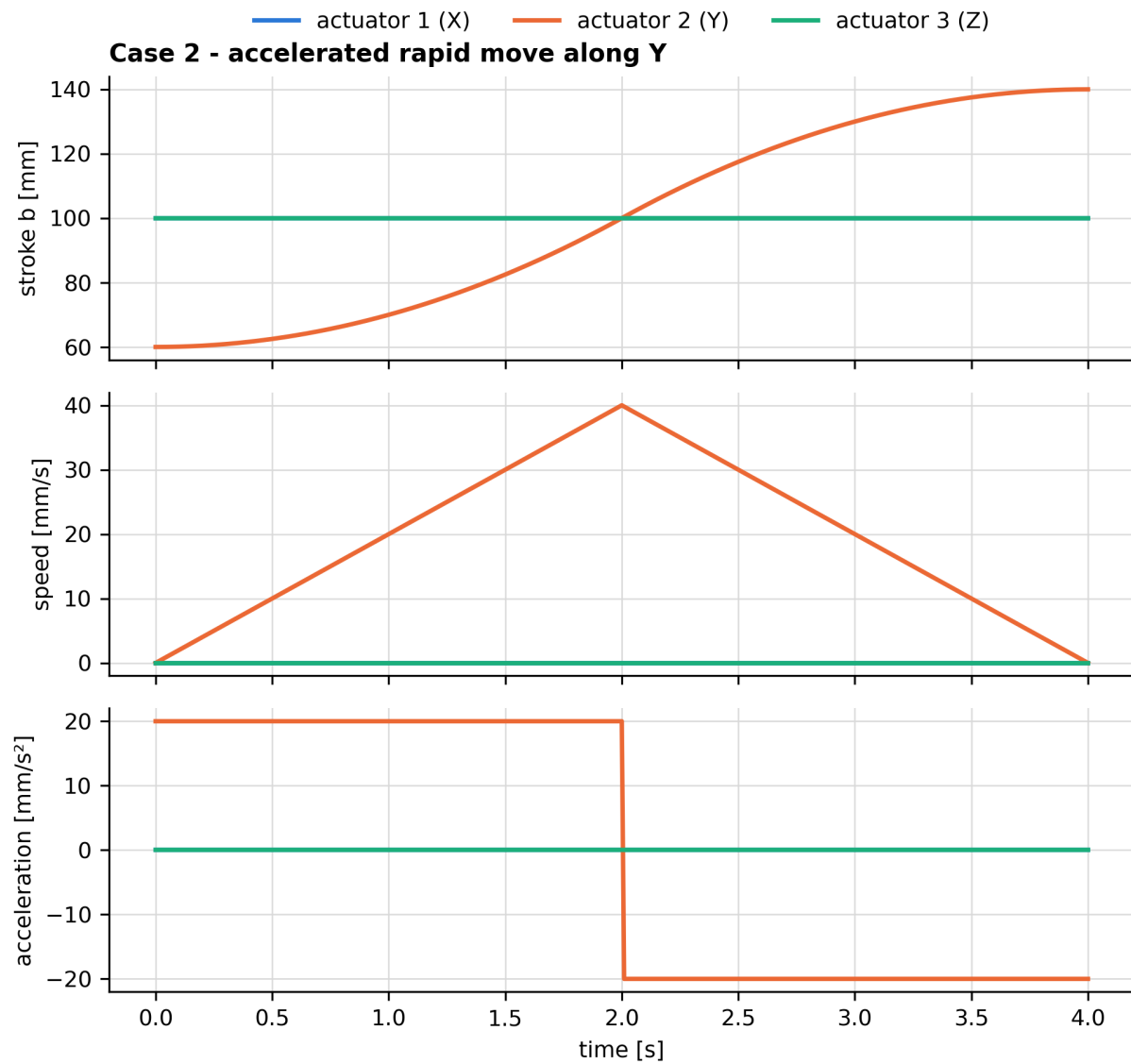


Figure 2: Case 2 – accelerated rapid traverse along y .

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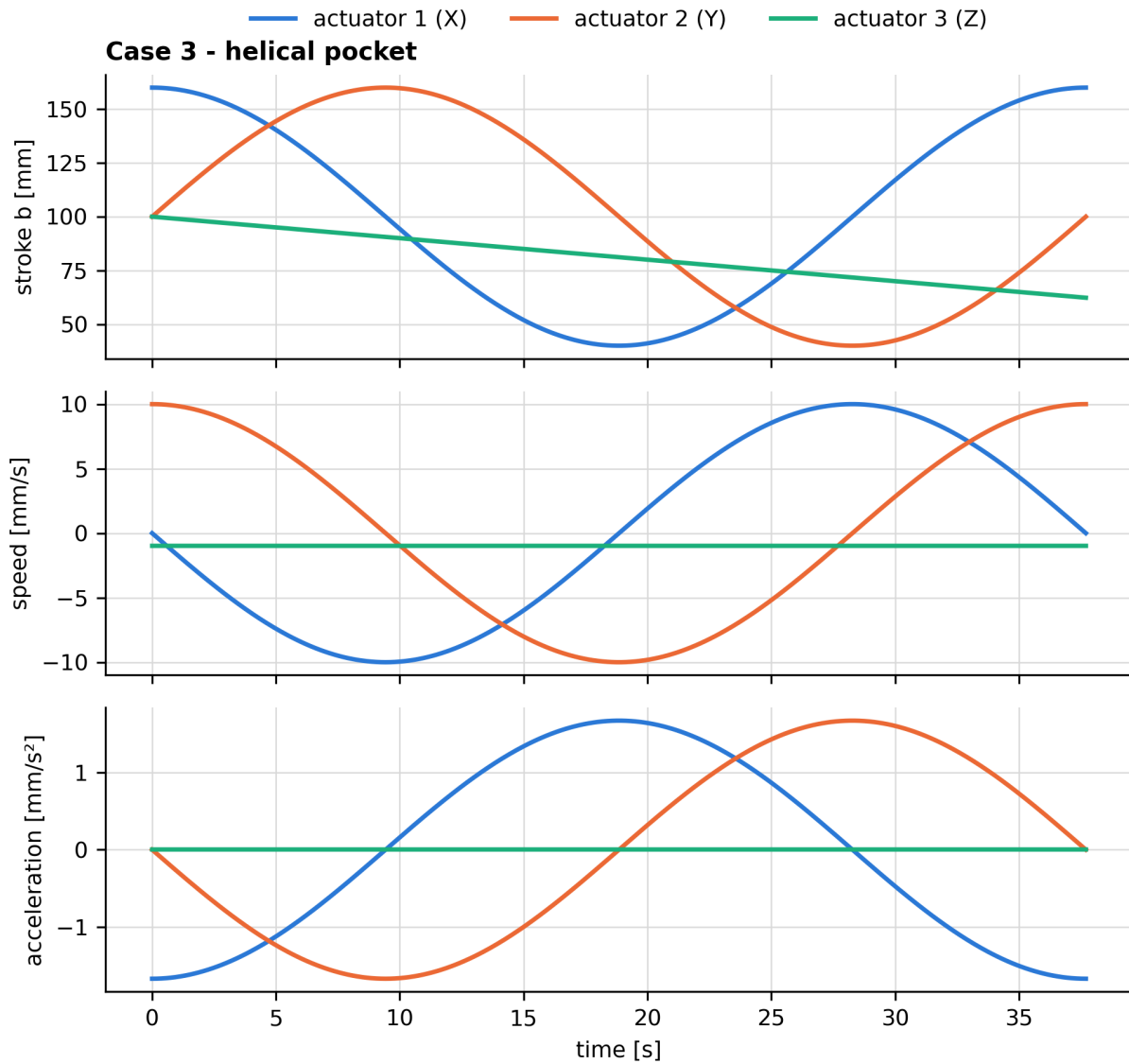


Figure 3: Case 3 – helical pocketing move.

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Workspace

The reachable workspace was mapped by solving the inverse kinematics on a 5 mm grid and rejecting the poses for which any leg leaves its stroke or any 2R chain becomes unreachable ($r > \ell_C + \ell_D$ or $r < |\ell_C - \ell_D|$). Out of 68,921 sampled points, the full 200 mm cube is reachable:

Quantity	Value
Bounds in x, y, z	± 100 mm each
Usable volume	8.615 dm^3
Shape	Cube, no interior singularity

The workspace is a perfect cube — a direct consequence of the decoupling — and no singular pose was found inside it, since $\det \mathbf{J}$ keeps a constant sign over the whole grid. This is the main practical advantage of the Tripteron over a delta or a Stewart platform of comparable size.

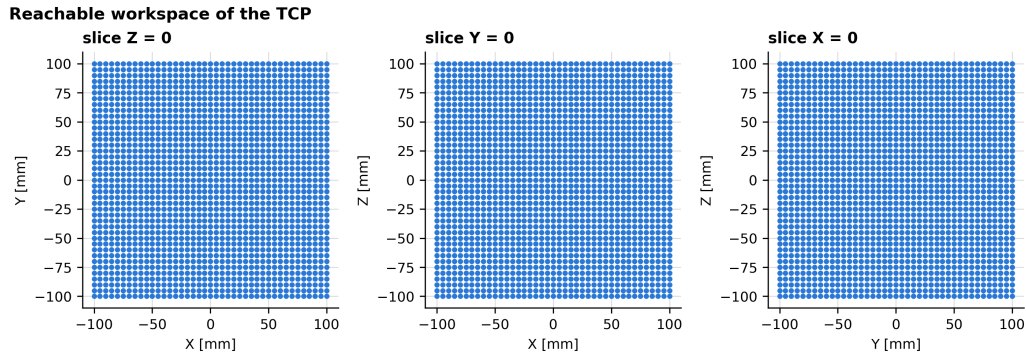


Figure 4: Reachable workspace of the machine: the full 200 mm cube.

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Description

With the geometry of every pose known from Sheet 1, the second calculation determines the forces that the machine has to carry: the axial force each lead screw must produce, the reactions at the twelve joints, and the internal bending moment of the six arms. The load case is the cutting force at the tool tip plus the self weight of all moving bodies.

The Tripteron is an *over-constrained* parallel mechanism: three closed loops share a single platform. The rigid-body equilibrium system is therefore statically indeterminate, and this is treated explicitly below instead of being ignored.

Objective of the calculation

1. Obtain the axial force on each actuator for the worst pose inside the workspace.
2. Obtain the joint reactions and the maximum bending stress in the arms.
3. Define the design axial load that Sheet 3 uses to size the lead screw.
4. Cross-check the result with an independent energy method.

Variables of the calculation model

Each of the ten moving bodies (three carriages, six links and the platform) contributes six equilibrium equations, and each joint contributes its unknown reaction components: six per prismatic joint (two transverse forces, three moments and the actuator force) and five per revolute joint (three forces and the two moments perpendicular to its axis), i.e. 21 per leg. Assembling the whole machine gives

Equilibrium equations	60
Unknown reaction components	63
Rank of the system matrix	60
Redundant (statically indeterminate) components	3

The three redundant components are the price of the closed kinematic chains: the platform is held by three legs where two would already fix its three translations. A rigid-body model cannot distribute those three components; only an elastic model can.

This number is not arbitrary. The mobility count of Delivery 1 gives, for 10 moving bodies and 12 one-degree-of-freedom joints, $M = 6 \times 10 - 12 \times 5 = 0$, while the machine evidently has three degrees of freedom. The difference of three is precisely the number of redundant constraints introduced by the special geometry of the Tripteron, and it reappears here as the three redundant reaction components. The kinematic and the static models therefore agree on the degree of over-constraint of the machine.

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Calculation model

Rigid-body equilibrium

For every body k with weight $\mathbf{W}_k$ acting at its centre of mass $\mathbf{c}_k$,

$$\sum_j \mathbf{R}_{kj} + \mathbf{W}_k + \mathbf{F}_{\text{ext},k} = \mathbf{0}, \quad \sum_j (\mathbf{r}_{kj} - \mathbf{c}_k) \times \mathbf{R}_{kj} + \mathbf{M}_{\text{ext},k} = \mathbf{0} \quad (12)$$

where $\mathbf{R}_{kj}$ is the reaction transmitted to body k at joint j and $\mathbf{r}_{kj}$ is the position of that joint. Each revolute joint transmits three force components and the two moment components perpendicular to its axis; each prismatic joint transmits the two force components perpendicular to its axis, the three moment components, and the actuator force along the axis.

Collecting the unknowns in $\mathbf{x} \in \mathbb{R}^{63}$ gives the linear system $\mathbf{A}\mathbf{x} = \mathbf{y}$ with $\mathbf{A} \in \mathbb{R}^{60 \times 63}$. Since $\text{rank}(\mathbf{A}) = 60$ the system is consistent but has a three-parameter family of solutions. The *minimum-norm* solution is selected,

$$\mathbf{x}^* = \arg \min \|\mathbf{x}\|_2 \quad \text{subject to} \quad \mathbf{A}\mathbf{x} = \mathbf{y}, \quad (13)$$

computed with a least-squares (SVD) solver. This is the physically meaningful choice for a symmetric, equally stiff structure, and it is conservative for the arms because it does not concentrate the redundant load in a single leg.

Independent verification: virtual work

The actuator forces do not depend on the redundant components, so they can be obtained exactly from the principle of virtual work without solving the indeterminate system:

$$F_i = -\left(\mathbf{F}_c \cdot \frac{\partial \mathbf{P}}{\partial b_i} + \sum_k \mathbf{W}_k \cdot \frac{\partial \mathbf{c}_k}{\partial b_i}\right) \quad (14)$$

The derivatives $\partial \mathbf{c}_k / \partial b_i$ are evaluated by central finite differences of the position solution of Sheet 1, with a perturbation of 1×10^{-4} mm. This route shares nothing with the equilibrium assembly except the geometry, so agreement between the two is a genuine check.

Arm stresses

The arms are $25 \times 25 \times 2$ mm square tubes:

$$A = 184 \text{ mm}^2, \quad I = 16,345 \text{ mm}^4, \quad c = 12.5 \text{ mm}, \quad W = \frac{I}{c} = 1307.6 \text{ mm}^3 \quad (15)$$

and the bending stress follows from the internal moment at the built-in end, $\sigma_b = M/W$.

Values of the calculation model

The cutting force is $\mathbf{F}_c = [80, 80, -120]^T$ N: two horizontal components from the end-mill and a vertical component pointing *downwards*, i.e. pressing the tool into the part. Four poses were evaluated: the home

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pose and the three corners of the workspace that maximise, respectively, the vertical, the diagonal and the mixed load.

Results and conclusions

Actuator forces

Pose	F_x [N]	F_y [N]	F_z [N]	Worst arm	σ_b [MPa]
Home (0, 0, 0)	−80.9	−80.9	+157.9	link C3	25.4
Corner (+90, +90, +90)	−81.4	−81.4	+156.3	link C3	25.1
Corner (−90, −90, −90)	−79.4	−79.4	+162.0	link C3	26.1
Corner (+90, −90, +90)	−81.4	−79.4	+155.0	link C3	24.9

The equilibrium residual $\|\mathbf{Ax} - \mathbf{y}\|$ is below 2.7×10^{-9} N in every pose, and the actuator forces obtained from equation (14) agree with the equilibrium solution to better than 1×10^{-6} N. Two completely independent methods therefore give the same answer.

Design axial load: $F_{\max} = 162.0$ N, at the corner (−90, −90, −90), on the vertical axis. This value is carried forward to Sheet 3.

Interpretation

- The horizontal actuators carry almost exactly the horizontal cutting force (≈ 81 N against 80 N applied); the extra 0.9 N is the weight of the arms projected on the horizontal axes. This is the expected behaviour of a decoupled machine and is a further confirmation of the kinematic model.
- The vertical actuator carries the vertical cutting force *plus* the entire weight of the moving assembly: with the spindle off, gravity alone demands 37.9 N on axis z against 0.9 N on each horizontal axis. The vertical screw is therefore the critical one and must never be back-driven.
- The variation of the actuator force across the whole workspace is only 4.5 % (155.0 to 162.0 N), so a single worst-case load is a sound basis for sizing the transmission.
- The maximum bending stress in the arms is 26.1 MPa against a yield strength of 250 MPa for AISI 1020, i.e. a safety factor of **9.6**. The arms are not the limiting component; they were sized by stiffness and by the need to accommodate the bearing bores, not by strength.

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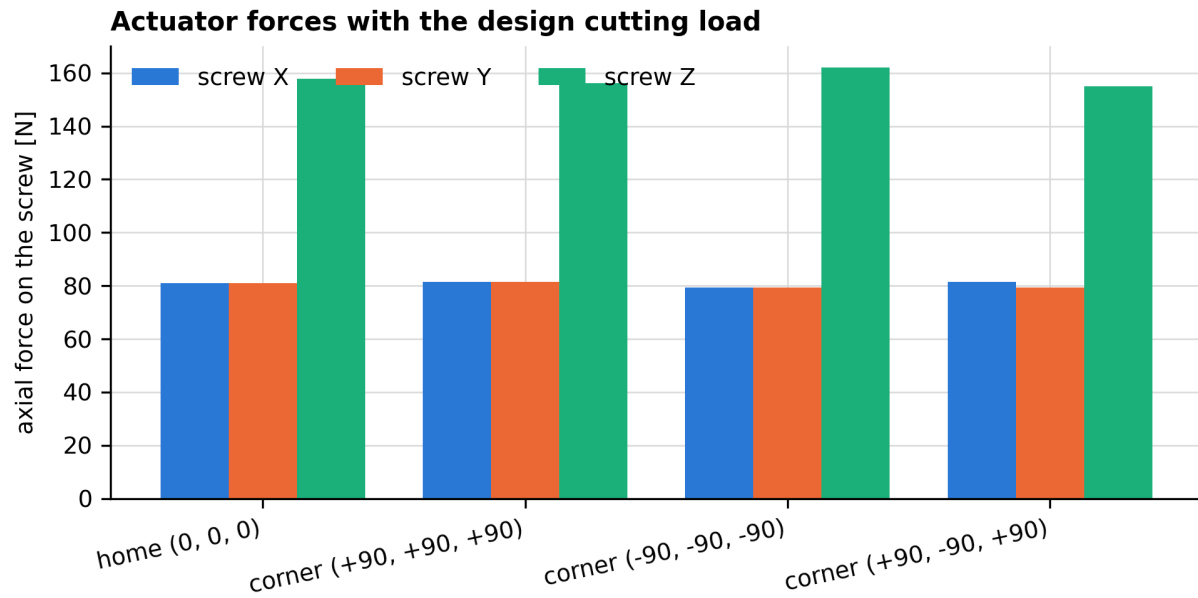


Figure 5: Actuator forces and worst arm bending stress for the four poses analysed.

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Description

Each axis is driven by a trapezoidal lead screw turned directly by a stepper motor. The nut is fixed to the carriage, which runs on two linear guides (the R-pair guide rails of Delivery 2) that take all the transverse load, so the screw is loaded axially and in torsion only. This sheet sizes that screw for the design load obtained in Sheet 2.

Objective of the calculation

1. Compute the torque required to raise and to lower the load, including collar friction.
2. Compute the mechanical efficiency and check whether the screw is self-locking.
3. Verify the screw body against the combined axial, torsional, thread-shear and bearing stresses.
4. Verify the screw as a column in compression.

Variables of the calculation model

Symbol	Description	Value
d	Nominal diameter	8 mm
p	Thread pitch	2 mm
n_s	Number of starts	4
L	Lead, $L = n_s p$	8 mm
d_m	Mean diameter, $d - p/2$	7 mm
d_r	Root diameter, $d - p$	6 mm
α	Half thread angle (trapezoidal)	15°
λ	Lead angle, $\arctan(L/\pi d_m)$	19.99°
μ	Thread friction coefficient (steel–bronze, greased)	0.15
μ_c	Collar friction coefficient (thrust bearing)	0.02
d_c	Mean collar diameter	6 mm
n_t	Engaged threads in the nut	6
ℓ	Unsupported screw length	500 mm
E	Young's modulus, steel	207 GPa
S_y	Yield strength	370 MPa
F	Design axial load (Sheet 2)	162.0 N

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Calculation model

Torque (Shigley, power screws)

For a trapezoidal thread the friction coefficient is divided by $\cos \alpha$:

$$T_R = \frac{F d_m}{2} \left(\frac{L + \pi \mu d_m \sec \alpha}{\pi d_m - \mu L \sec \alpha} \right) + \frac{F \mu_c d_c}{2} \quad (16)$$

$$T_L = \frac{F d_m}{2} \left(\frac{\pi \mu d_m \sec \alpha - L}{\pi d_m + \mu L \sec \alpha} \right) + \frac{F \mu_c d_c}{2} \quad (17)$$

and the efficiency of the screw pair is

$$e = \frac{F L}{2\pi T_R}. \quad (18)$$

Equation (18) is bounded above by unity by construction. *Any result above 1 means an arithmetic error, not a good screw.*

Self-locking

The screw holds the load without a brake only if the thread friction torque exceeds the torque produced by the load, i.e. if the first term of (17) is positive:

$$\mu \geq \frac{L \cos \alpha}{\pi d_m}. \quad (19)$$

Stresses in the screw body

With $A_r = \pi d_r^2/4$ the root area, n_t engaged threads and thread thickness $p/2$ at the root:

$$\sigma_a = \frac{F}{A_r} \quad \tau_t = \frac{16 T_R}{\pi d_r^3} \quad (20)$$

$$\tau_{\text{thread}} = \frac{2F}{\pi d_r p n_t} \quad \sigma_{\text{bearing}} = \frac{2F}{\pi d_m p n_t} \quad (21)$$

and the distortion-energy (von Mises) equivalent stress is

$$\sigma' = \sqrt{\sigma_a^2 + 3(\tau_t^2 + \tau_{\text{thread}}^2)}, \quad n = \frac{S_y}{\sigma'}. \quad (22)$$

Column buckling

The screw is treated as a pinned–pinned column, $K = 1$. With $I = \pi d_r^4/64$, $r_g = \sqrt{I/A_r} = d_r/4$ and slenderness ℓ_e/r_g , the transition slenderness between the Johnson and the Euler regimes is

$$\left(\frac{\ell_e}{r_g} \right)_1 = \sqrt{\frac{2\pi^2 E}{S_y}}, \quad (23)$$

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and the critical load is

$$P_{cr} = \frac{\pi^2 EI}{(K\ell)^2} \quad (\text{Euler}), \quad P_{cr} = A_r \left[S_y - \frac{1}{E} \left(\frac{S_y \ell_e}{2\pi r_g} \right)^2 \right] \quad (\text{Johnson}). \quad (24)$$

Values and results

Torque and efficiency

Quantity	Design load, 162.0 N	Delivery 1 load, 151.9 N
Thread torque, raising $T_{R,\text{thread}}$	311.9 N mm	292.5 N mm
Collar torque T_c	9.7 N mm	9.1 N mm
Total raising torque T_R	321.6 N mm	301.6 N mm
Total lowering torque T_L	−102.2 N mm	−95.8 N mm
Efficiency e	64.1 %	64.1 %
μ required for self-locking	0.351	0.351
Self-locking with $\mu = 0.15$?	No	No
Fraction of motor holding torque used	7.1 %	6.7 %

Stresses and safety factors

Quantity	Value	Limit / remark
Root area A_r	28.27 mm ²	—
Axial stress σ_a	5.73 MPa	compression
Torsional stress τ_t	7.58 MPa	from T_R
Thread shear τ_{thread}	3.77 MPa	6 engaged threads
Bearing stress σ_{bearing}	1.23 MPa	bronze nut, limit ≈ 17 MPa
von Mises σ'	14.33 MPa	$S_y = 370$ MPa
Static safety factor	25.8	—
Second moment I	63.6 mm ⁴	$\pi d_r^4/64$
Radius of gyration r_g	1.5 mm	$d_r/4$
Slenderness ℓ/r_g	333	limit 105 $\Rightarrow$ Euler
Critical load P_{cr}	519.9 N	pinned–pinned
Buckling safety factor	3.21	—

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Calculation	Sheet 3 – Lead-screw drive	Date	September 2026
Analyst	D. Amell / G. Riaño / D. Zuluaga	Reviewer	Juan Andrés Gallego Sánchez

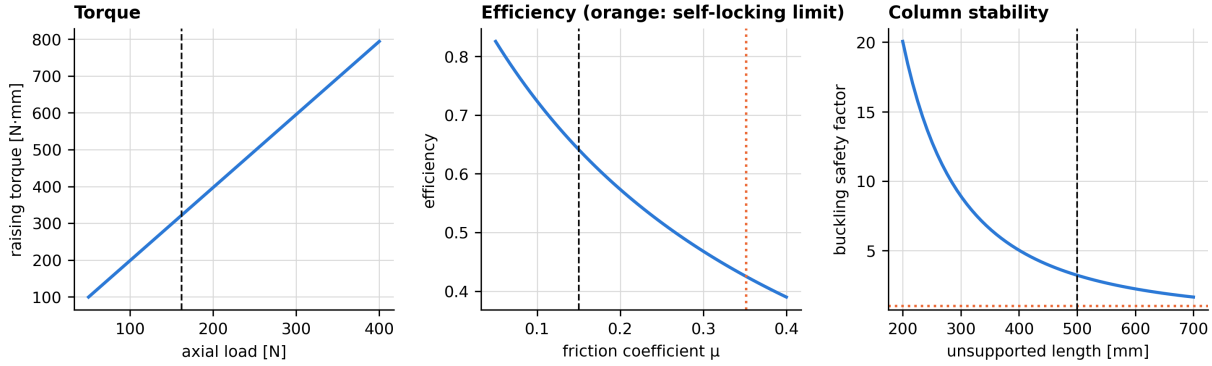


Figure 6: Lead-screw verification: torque breakdown, stress components and column check.

Conclusions

- Strength is not the driver.** The von Mises stress of 14.3 MPa gives a safety factor of 25.8. The 8 mm screw is far stronger than the load requires; it was selected for availability and for the compatibility of its 8 mm lead with the required feed rates.
- Buckling governs the screw.** With a safety factor of 3.21 against Euler buckling, the column check is eight times more restrictive than the stress check. This is the correct behaviour for a long, slender screw and it is the criterion that must be re-examined if the stroke is ever increased: P_{cr} falls with the square of the length, so a 700 mm screw would drop to 265 N and a safety factor of 1.6, which would no longer be acceptable.
- The screw is not self-locking.** A 4-start thread has a lead angle of 19.99° ; equation (19) requires $\mu \geq 0.351$ while the greased steel–bronze pair gives $\mu = 0.15$. The vertical axis *will* back-drive under gravity if the motor is de-energised. Two consequences, both already covered by the selected hardware:
 - the closed-loop NEMA 34 motor must keep holding current at standstill — it needs only 0.32 N m out of 4.5 N m, i.e. 7.1 % of its holding torque;
 - an electromagnetic brake or a mechanical stop is mandatory on the z axis for maintenance and for a power failure. This is a safety item, not an optional one.

The efficiency of 64.1 % is the direct trade-off: a fast, efficient, non-self-locking screw.

- Motor sizing is comfortable.** 0.32 N m required against 4.5 N m available leaves a factor of 14 for acceleration, preload, friction ageing and the inertia of the screw itself.

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Calculation	Sheet 4 – Chassis finite element verification	Date	September 2026
Analyst	D. Amell / G. Riaño / D. Zuluaga	Reviewer	Juan Andrés Gallego Sánchez

Description

The three legs are mounted on a welded frame of square steel tubing that also carries the work table. The frame is the component whose behaviour cannot be captured by a hand calculation, because it is a three-dimensional redundant structure, so it was verified by finite elements in SolidWorks Simulation. This sheet reports and interprets those results; the raw nodal data exported from the solver are stored in `04_Calculations/data/fea_exports/` and are post-processed by the same Python package that produced the rest of this report, so the numbers below are not transcribed by hand.

Objective of the calculation

1. Verify that the frame stays in the elastic range under the reactions computed in Sheet 2.
2. Verify that its deflection is small compared with the machining tolerance.
3. Verify the frame against global buckling.

Variables and model

Element type	Beam elements (welded tubular frame)
Material	Alloy steel, $E = 210 \text{ GPa}$, $S_y = 620 \text{ MPa}$
Boundary condition	Four feet fixed to the floor
Loads	Joint reactions from Sheet 2 plus self weight
Studies	Linear static + linear (eigenvalue) buckling

Beam elements are appropriate here: the frame is made of slender tubes loaded mainly in bending and axial force, welded at the nodes, and a beam idealisation reproduces that behaviour with a fraction of the degrees of freedom of a solid mesh. Local weld stresses are not captured by this model and are covered by the weld sizing of Delivery 2.

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Calculation	Sheet 4 – Chassis finite element verification	Date	September 2026
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Results

Quantity	Samples	Minimum	Maximum	$ \cdot _{\max}$
Bending stress [MPa]	19	−9.36	−0.574	9.36
Axial stress [MPa]	33	−0.305	0.0029	0.305
Shear stress, dir. 1 [MPa]	40	−0.987	2.69	2.69
Shear stress, dir. 2 [MPa]	31	−0.219	0.627	0.627
Strain [−]	23	-6.4×10^{-6}	2.9×10^{-6}	6.4×10^{-6}
Displacement [mm]	22	0	0.1016	0.1016
Buckling mode 1, x [mm]	27	0	0.0285	0.0285
Buckling mode 1, y [mm]	24	−0.0035	0.0035	0.0035
Buckling mode 1, z [mm]	21	−0.0213	0.0019	0.0213

Verification	Result
Governing stress	bending, 9.36 MPa
Yield strength	620 MPa
Static safety factor	66
Maximum displacement	0.102 mm
Buckling load factor	2238

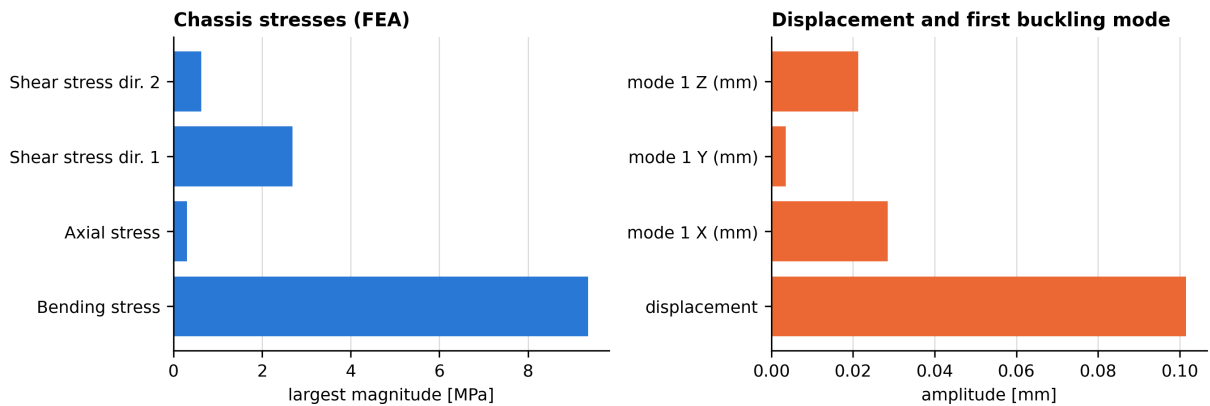


Figure 7: Finite element results: range of each exported quantity and resulting margins.

Conclusions

1. **The frame is massively over-strength.** A safety factor of 66 against yield and a buckling load factor of 2238 mean the structure is nowhere near its strength limit. This is not an error: the frame was sized by *stiffness* and by the commercial tube sizes available, not by stress. In a machine tool the deflection under load is what shows up on the part, and it is always the binding requirement.

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Calculation	Sheet 4 – Chassis finite element verification	Date	September 2026
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2. **Stiffness is the meaningful result.** The maximum displacement of 0.102mm occurs at the top rail, furthest from the feet. Referred to the 200mm workspace this is 0.05%, and it is of the same order as the positioning resolution of the drives. It is acceptable for the intended use (light milling of aluminium and plastics) but it is the single number that limits the achievable accuracy of the machine, so any future increase of the cutting force should be checked against it first.
3. **The dominant stress is bending,** an order of magnitude above the axial and shear components. The load path is therefore what one would expect: the reactions of the legs are applied off the neutral axis of the rails and are carried in bending to the feet. If the frame were ever to be lightened, the rails in bending are what must be preserved; the diagonal members carry little load.
4. **Global buckling is irrelevant** for this structure at this load level. The first mode is a lateral sway of the uprights, and it would require 2238 times the service load to occur.

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Calculation	Appendix A – Corrections	Date	September 2026
Analyst	D. Amell / G. Riaño / D. Zuluaga	Reviewer	Juan Andrés Gallego Sánchez

Corrections to the previous calculation memo

The earlier version of this memo contained a number of results that cannot be correct on physical grounds. Each of them is listed below with the reason it is wrong and the value that replaces it. All the replacement values come from the recomputation documented in Sheets 1–4.

#	Previous result	Why it cannot be right	Corrected value
1	Screw efficiency reported above 100 %	Efficiency is the ratio of useful work to input work; a value above 1 would create energy. The error came from inverting the numerator and denominator of equation (18).	$e = FL/(2\pi T_R) = 64.1 \%$
2	Thread shear stress 2379.96 MPa	Six times the yield strength of the material. The shear area was taken as a single thread root cross-section instead of the cylindrical surface of the engaged threads.	$\tau = 2F/(\pi d_r p n_t) = 3.77 \text{ MPa}$
3	Critical load written as $P_{cr} = 2EI/\ell^2$ with $E = 20,000 \text{ MPa}$	The Euler formula is $\pi^2 EI/(K\ell)^2$, and the modulus of steel is 207,000 MPa, not 20,000 MPa. Two independent errors in the same expression.	$P_{cr} = \pi^2 EI/(K\ell)^2 = 519.9 \text{ N}$, $n = 3.21$
4	Slenderness ratio inconsistent with the regime used	The radius of gyration was taken as $d_r/2$ instead of $d_r/4$, which placed the screw in the Johnson regime when it is a long Euler column.	$\ell/r_g = 333$ against a transition value of 105 $\Rightarrow$ Euler regime
5	Two different axial stresses for the same screw, 3.438 MPa and 72.21 MPa	The two values differ by a factor of 21; the second used the thread-root <i>area</i> where a diameter belonged.	$\sigma_a = F/A_r = 5.73 \text{ MPa}$

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Calculation	Appendix A – Corrections	Date	September 2026
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#	Previous result	Why it cannot be right	Corrected value
6	Static analysis returning zero reactions at every joint	The trivial solution of a homogeneous system. The equilibrium matrix had been assembled without the external load vector, so the solver returned $\mathbf{x} = \mathbf{0}$.	60 equations, 63 unknowns, rank 60; minimum-norm solution, residual $< 3 \times 10^{-9}$ N, cross-checked by virtual work
7	Pitch, lead and diameters used interchangeably	With 4 starts the lead is four times the pitch; using p where L belongs changes the lead angle from 19.99° to 5.2° and reverses the self-locking conclusion.	$p = 2$ mm, $L = n_s p = 8$ mm, $d_m = 7$ mm, $d_r = 6$ mm
8	Cutting force applied upwards	An end mill pushes the work downwards against the table; applied upwards it partially cancels the weight of the moving assembly and under-predicts the load on the vertical screw.	$\mathbf{F}_c = [80, 80, -120]^\top$ N $\Rightarrow$ design load 162.0 N instead of 85 N
9	Self-locking claimed for the vertical axis	Consequence of #7. A 4-start screw with $\mu = 0.15$ cannot be self-locking.	Not self-locking; a brake or permanent holding current is required on z
10	No verification of the kinematic solution	A position solution with no residual and no independent check cannot be audited.	Newton–Raphson residual $< 8 \times 10^{-11}$ mm, agreement with the closed-form inverse kinematics to 10^{-13} mm, velocities and accelerations checked against finite differences

Effect of the corrections on the design

None of the corrections invalidates the machine. The two that matter in practice are:

- **Correction 8** raises the design axial load from 85 N to 162.0 N. The screw still passes every check, but the buckling safety factor drops from about 6 to 3.21, which is why the column check — and not the stress check — is now the governing criterion for the transmission.
- **Correction 9** changes an operating requirement: the vertical axis is not self-locking and needs a brake or a permanently energised motor. This is a safety item that the previous memo did not identify.

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Calculation	Appendix B – Calculation software	Date	September 2026
Analyst	D. Amell / G. Riaño / D. Zuluaga	Reviewer	Juan Andrés Gallego Sánchez

Reproducing the results

Every figure and every number in this report is generated by the Python package stored in the folder `04_Calculations`. A single command regenerates all of them:

`python run_all.py`

The script creates its own virtual environment on first use, installs the dependencies and writes the results to `results/` (CSV and JSON) and `figures/` (PNG).

Module	Contents
<code>parameters.py</code>	All machine data in one place: geometry, masses, materials, loads.
<code>kinematics.py</code>	Constraints, Jacobian, $\dot{\mathbf{J}}$, Newton–Raphson solver, closed-form inverse kinematics, trajectory generation, workspace mapping.
<code>statics.py</code>	Assembly of the 60×63 equilibrium system, minimum-norm solution, virtual-work actuator forces, arm section properties and bending stresses.
<code>power_screw.py</code>	Screw geometry, raising/lowering torque, efficiency, self-locking, stresses, Euler/Johnson column check.
<code>fea_postprocess.py</code>	Reads the nodal data exported from SolidWorks Simulation and computes the ranges and safety factors of Sheet 4.
<code>run_all.py</code>	Runs the four studies and writes results and figures.

Verification strategy

Each physical result is produced by two independent routes, and the report states the discrepancy between them:

Result	Method 1	Method 2
Position	Newton–Raphson on $\Phi(\mathbf{q}, t) = 0$	Closed-form 2R inverse kinematics
Velocity, acceleration	Analytical $\mathbf{J}$, $\dot{\mathbf{J}}$	Finite differences of the position solution
Actuator forces	60×63 rigid-body equilibrium	Principle of virtual work
Screw	Shigley closed-form expressions	Dimensional and physical bounds ($e \leq 1$, $\sigma < S_y$)
Chassis	Finite elements	Order-of-magnitude hand check of the rails in bending